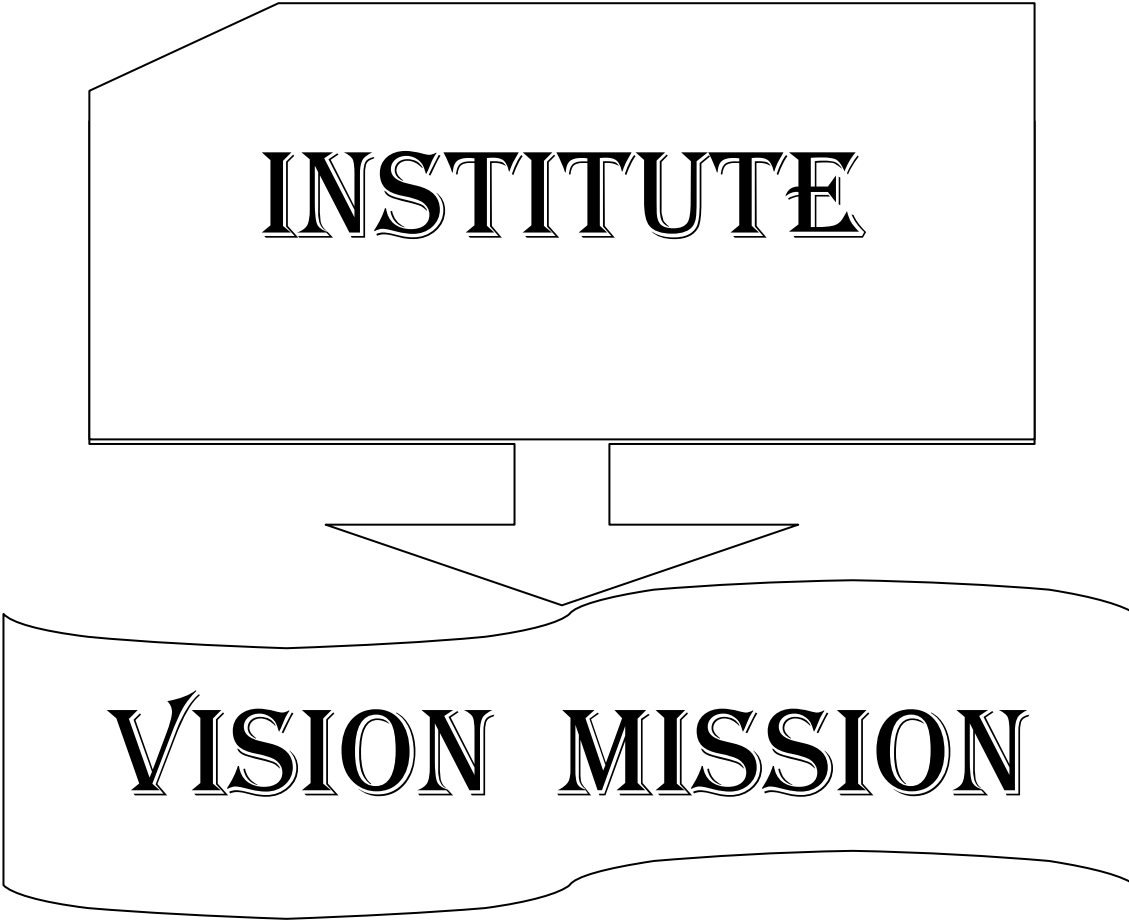


Academic Year
2021-22
V SEM CSE & CST
Handbook
(V18 Regulation)

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INSTITUTE

VISION MISSION

INSTITUTE VISION AND MISSION

VISION

To be a premier technological institute striving for excellence with global perspective and commitment to the nation.

MISSION

- To produce engineering graduates of professional quality and global perspective through Learner Centric Education.
- To establish linkages with government, industry and research laboratories to promote R&D activities and to disseminate innovations.
- To create an eco-system in the institute that leads to holistic development and ability for life-long learning.

DEPARTMENT

VISION

MISSION

DEPARTMENT

VISION AND MISSION

Vision:

- To evolve as a centre of academic and research excellence in the area of Computer Science and Engineering.

Mission :

- To utilize innovative learning methods for academic improvement.
- To encourage higher studies and research to meet the futuristic requirements of Computer Science and Engineering.
- To inculcate Ethics and Human values for developing students with good character

**PROGRAM
EDUCATIONAL
OBJECTIVES,
PROGRAM OUTCOMES
& PROGRAM
SPECIFIC
OUTCOMES**

Program Educational Objectives (PEOs)

Graduates of this programme will :

PEO 1: Adapt to evolving technology.

PEO 2: Provide optimal solutions to real time problems.

PEO 3: Demonstrate his/her abilities to support service activities with due consideration for Professional and Ethical Values.

Programme Specific Outcomes (PSO s):

A graduate of the Computer Science and Engineering Program will be able to:

PSO 1: Use Mathematical Abstractions and Algorithmic Design along with Open Source Programming tools to solve complexities involved in Programming. [K3]

PSO 2: Use Professional engineering practices and strategies for development and maintenance of software. [K3]

Program Outcomes (POs):

Computer Science Engineering Graduates will be able to:

1. **Engineering knowledge:** Apply the knowledge of Mathematics, Science, Engineering Fundamentals and Concepts of Computer Science Engineering to the solution of complex Engineering problems. [K3]
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of Mathematics, Natural Sciences and Computer Science. [K4]
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specific needs with appropriate consideration for the public health and safety, and the cultural, societal and environmental considerations. [K5]
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions. [K5]
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex Engineering activities with an understanding of the limitations. [K3]
6. **The Engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional Engineering practice. [K3]
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for, sustainable development. [K3]
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the Engineering practice. [K3]
9. **Individual and teamwork:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings. [K6]
10. **Communication:** Communicate effectively on complex Engineering activities with the Engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions. [K2]
11. **Project management and finance:** Demonstrate knowledge and understanding of the Engineering and Management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments. [K6]
12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change. [K1]

ACADEMIC CALENDAR

✉ : principal@srivasaviengg.ac.in
svec.a8@gmail.com



☎ : 08818- 284344, 355

SRI VASAVI ENGINEERING COLLEGE (AUTONOMOUS)

(Sponsored by Sri Vasavi Educational Society)

(Approved by AICTE, New Delhi & Permanently affiliated to JNTUK, Kakinada)

(Accredited by NAAC with 'A' Grade, Recognized by UGC under section 2(f) & 12(B))

(NBA Accreditation to B.Tech., EEE, CSE, ME and ECE Branches for 3 Years)

Pedatadepalli, **TADEPALLIGUDEM** – 534 101. W.G.Dist. (A.P)

Principal's Office
Date: 04-09-2021

Academic Calendar **For B.Tech V sem & VI sem , Academic Year 2021-22** **(2019 Admitted Batch)**

V Semester			
Description	From	To	Weeks
Commencement of Class Work	15.09.2021		
I Unit of Instructions	15.09.2021	30.10.2021	7 W
I Mid Examinations	01.11.2021	06.11.2021	1 W
II Unit of Instructions	08.11.2021	25.12.2021	7 W
II Mid Examinations	27.12.2021	01.01.2022	1 W
Preparation & Practicals	03.01.2022	08.01.2022	1 W
End Examinations	10.01.2022	22.01.2022	2 W
Commencement of VI Semester Class Work	31.01.2022		
VI Semester			
I Unit of Instructions	31.01.2022	19.03.2022	7 W
I Mid Examinations	21.03.2022	26.03.2022	1 W
II Unit of Instructions	28.03.2022	14.05.2022	7 W
II Mid Examinations	16.05.2022	21.05.2022	1 W
Preparation & Practicals	23.05.2022	28.05.2022	1 W
End Examinations	30.05.2022	11.06.2022	2 W
Commencement of Next Year Class Work			
Note : Calendar is prepared with 8 hrs/day hence 7 weeks per instruction period. Internal Examinations shall be conducted during the instructional days.			


PRINCIPAL

Copy to : ALL

Vision

To be a premier technological institute striving for excellence with global perspective and commitment to the nation.

Mission

- To produce Engineering graduates of professional quality and global perspective through learner-centric education.
- To establish linkages with government, industry and Research laboratories to promote R&D activities and to disseminate innovations.
- To create an eco-system in the institute that leads to holistic development and ability for life-long learning.



SRI VASAVI ENGINEERING COLLEGE (Autonomous)

Pedatadepalli, TADEPALLIGUDEM-534 101, W.G. Dist.

Department Of Computer Science & Engineering (Accredited by NBA)



CLASS CONSOLIDATED TIME TABLE

Class: V SEM Section – A Class Coordinator: Mrs. Y DivyaVani Room: G-202 w.e.f. 15-09-2021

Periods	1	2	3	4	1:00P M 2:00P M	5	6	7
Time	(09.30 AM- 10.30 AM)	(10.30 AM- 11.20 AM)	(11.20 AM- 12.10 PM)	(12.10 PM- 01.00 PM)		(02.00 PM- 02.50 PM)	(02.50 PM- 03.40 PM)	(03.40 PM- 04.30 PM)
Day								
Mon	DBMS	PCS-III (Aptitude)		UNP	LUNCH BREAK	PCS-III (Verbal)		AI
Tue	CN	UNP	DAA	DBMS		TS-III		
Wed	OS	DBMS Lab				AI	OB	DBMS
Thu	AI	UNP	OB	OS		DBMS	CN	OS
Fri	DAA	TS-III				CN	OS	DAA
Sat	OB	DAA	CN	AI		OS & UNIX Lab		

Class: V SEM Section – B Class Coordinator: Mr. J Chandra Sekhar Room: G-203 w.e.f. 15-09-2021

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time	(09.30 AM- 10.30 AM)	(10.30 AM- 11.20 AM)	(11.20 AM- 12.10 PM)	(12.10 PM- 01.00 PM)		(02.00 PM- 02.50 PM)	(02.50 PM- 03.40 PM)	(03.40 PM- 04.30 PM)
Day								
Mon	PCS-III (Verbal)		DAA	CN	LUNCH BREAK	PCS-III (Aptitude)		DBMS
Tue	AI	OS & UNIX Lab				DAA	CN	OS
Wed	DAA	TS-III				CN	OS	OB
Thu	OS	DBMS	OB	UNP		DAA	AI	UNP
Fri	CN	AI	DBMS	OB		DBMS Lab		
Sat	DBMS	AI	OS	UNP		TS-III		

Class: V SEM Section – C Class Coordinator: Ms. D SasiRekha Room: G-204 w.e.f. 15-09-2021

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time	(09.30 AM- 10.30 AM)	(10.30 AM- 11.20 AM)	(11.20 AM- 12.10 PM)	(12.10 PM- 01.00 PM)		(02.00 PM- 02.50 PM)	(02.50 PM- 03.40 PM)	(03.40 PM- 04.30 PM)
Day								
Mon	DAA	OS	AI	DBMS	LUNCH BREAK	TS-III		
Tue	PCS-III (Aptitude)		OB	AI		PCS-III (Verbal)		OS
Wed	DBMS	CN	UNP	DAA		DBMS Lab		
Thu	CN	TS-III				DBMS	OB	DAA
Fri	OS	OS & UNIX Lab				AI	UNP	CN
Sat	UNP	AI	DBMS	DAA		OS	CN	OB

Class: V SEM Section – D Class Coordinator: Mr. K Venkatesh Room: G-205 w.e.f. 15-09-2021

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time Day	(09.30 AM- 10.30 AM)	(10.30 AM- 11.20 AM)	(11.20 AM- 12.10 PM)	(12.10 PM- 01.00 PM)		(02.00 PM- 02.50 PM)	(02.50 PM- 03.40 PM)	(03.40 PM- 04.30 PM)
Mon	DBMS	TS-III			LUNCH BREAK DBMS Lab AI DAA	DAA	CN	OS
Tue	OS	DBMS	PCS-III (Aptitude)			DBMS Lab		
Wed	PCS-III (Verbal)		AI	OS		DAA	UNP	CN
Thu	OB	DBMS	OS	AI		TS-III		
Fri	AI	OB	UNP	DAA		DBMS	CN	UNP
Sat	CN	OS & UNIX Lab				DAA	AI	OB

Class: V SEM Section – CST Class Coordinator: Mrs. P. Suneetha Room: B-203 w.e.f. 15-09-2021

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time	(09.30 AM- 10.30 AM)	(10.30 AM- 11.20 AM)	(11.20 AM- 12.10 PM)	(12.10 PM- 01.00 PM)		(02.00 PM- 02.50 PM)	(02.50 PM- 03.40 PM)	(03.40 PM- 04.30 PM)
Day								
Mon	DBMS	AI	DAA	OS	LUNCH BREAK	DBMS Lab		
Tue	PCS-III (Verbal)		DBMS	AI		TS-III		
Wed	AI	UNP	CN	OS		PCS-III (Aptitude)		DAA
Thu	CN	OS & UNIX Lab				DBMS	CN	OB
Fri	OS	TS-III				UNP	DAA	AI
Sat	UNP	DAA	DBMS	OB		CN	OS	OB

JH

Head of the Department



Staff Details:

S. No	Course Code	Course Name	Sections				
			A	B	C	D	CST
1.	V18CST10	Database Management Systems (DBMS)	Mr.M Yesu Shekharam	Mr.M Yesu Shekharam	Mr. K Venkatesh	Mr. K Venkatesh	Dr. P Laxmi Kanth
2.	V18CST11	Computer Networks (CN)	Mrs.D Anjani Suputri Devi	Mrs. D Anjani Suputri Devi	Mrs. D Anjani Suputri Devi	Mr. M S Kumar Reddy	Mr. M S Kumar Reddy
3.	V18CST12	Operating Systems (OS)	Mrs. Y DivyaVani	Mrs. Y DivyaVani	Ms. D SasiRekha	Ms. D SasiRekha	Mr. B Kiran Kumar
4.	V18CST13	Design and Analysis of Algorithms (DAA)	Mrs. M Anantha Lakshmi	Mrs. M Anantha Lakshmi	Mrs. M Anantha Lakshmi	Ms. M Madhavi	Ms. M Madhavi
5.	V18CST14	Unix Programming (UNP)	Dr.D Jaya Kumari	Mr. J N Chandra Sekhar	Mr. J N Chandra Sekhar	Mrs. P. Suneetha	Mrs. P. Suneetha
6.	V18CST17	Elective-I : Artificial Intelligence (AI)	Mr. K David Raju	Mr. K David Raju	Dr. G Loshma	Dr. G Loshma	Mr. K David Raju
7.	V18MBET53	Organizational Behavior (OB)	Mr.K Vinay Kumar	Ms.K LalithaBhavani	Mr. K Pavan Kumar	Ms. K Suji	Mr.P Bharath Kumar
8.	V18CSL06	Database Management Systems Lab (DBMS Lab)	Mr.M Yesu Shekharam / Ms. M Madhavi / Dr. P Laxmi Kanth	Mr.M Yesu Shekharam / Ms. M Madhavi / Mr. B Kiran Kumar	Mr. K Venkatesh / Mrs. M Anantha Lakshmi / Ms. D SasiRekha	Mr. K Venkatesh / Ms. M Madhavi / Mr.M Yesu Shekharam	Dr. P Laxmi Kanth / Mrs. D Anjani Suputri Devi / Mr. K Venkatesh
9.	V18CSL07	Operating Systems & Unix Lab (OS & UNIX Lab)	Mrs. Y DivyaVani / Dr.D Jaya Kumari / Mr. J N Chandra Sekhar	Mrs. Y DivyaVani / Mr. J N Chandra Sekhar / Mrs. P. Suneetha	Ms. D SasiRekha / Mr. J N Chandra Sekhar / Mrs. Y DivyaVani	Ms. D SasiRekha / Mrs. P. Suneetha / Mr. B Kiran Kumar	Mr. B Kiran Kumar / Mrs. P. Suneetha / Ms. M Madhavi
10.	V18ENT05	Professional Communication Skills -III (PCS-III) Aptitude	Mr.J N V Somayajulu	Mr.J N V Somayajulu	Mr.J N V Somayajulu	Mr. P Someswara Rao	Mr. P Someswara Rao
		Professional Communication Skills -III (PCS-III) English	Mr. T V S Suresh	Mr. T V S Suresh	Mr. T V S Suresh	Mr. G Srinivasa Rao	Mr. G Srinivasa Rao
11.	V18CST62	Technical Skills-III (TS-III)	Dr. V Venkateswara Rao	Dr. V Venkateswara Rao	Dr. V Venkateswara Rao	Dr. V Venkateswara Rao	Mrs.A Leelavathi

Head of the Department

Vision: To evolve as a centre of academic and research excellence in the area of Computer Science and Engineering.

Mission: To utilize innovative learning methods for academic improvement.
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To inculcate Ethics and Human values for developing students with good character

COURSE STRUCTURE

V - Semester							
S.No	Course Code		Course	L	T	P	C
1	V18CST10	PCC	Database Management Systems	3	0	0	3
2	V18CST11	PCC	Computer Networks	3	0	0	3
3	V18CST12	PCC	Operating Systems	3	0	0	3
4	V18CST13	PCC	Design and Analysis of Algorithms	3	0	0	3
5	V18CST14	PCC	Unix Programming	3	0	0	3
6	Elective – I						
	V18CST15	PEC	1.Advanced Computer Architecture	3	0	0	3
	V18CST16		2.Advanced Data Structures				
	V18CST17		3.Artificial Intelligence				
	V18CST18		4.Computer Graphics				
7	V18MBET53	HSS	Organizational Behavior	3	0	0	3
8	V18CSL06	PCC	Database Management Systems Lab	0	0	3	1.5
9	V18CSL07	PCC	Operating System and Unix Lab	0	0	3	1.5
10	V18ENT05		Professional Communication Skills -III	4	0	0	MNC
11	V18CST62		Technical Skills-III	4	0	0	MNC
Total				29	0	6	24



LESSON PLANS

Computer Networks

Academic Year: 2020-21

Semester: V

Name of the Course: Computer Networks

Programme: B.Tech

Section: A,B,C& D

Course Code:V18CST11/302

LESSON PLAN

Course Outcomes (Along with Knowledge Level):

S.No.	CO No.	Course Outcome	BTL
1.	1	Discuss fundamentals of network concepts and Reference Models	K2
2.	2	Discuss Communication media and switching techniques	K2
3.	3	Demonstrate Error control and protocols	K3
4.	4	Apply Routing algorithms and congestion control algorithms	K3
5.	5	Discuss Transport layer services and protocols	K2
6.	6	Describe Application layer protocols	K2

Text Books:

1. Computer Networks — Andrew S Tanenbaum, 4th Edition. Pearson Education/PHI
2. Data Communications and Networks – Behrouz A. Forouzan, Third Edition TMH

Reference Books:

1. An Engineering Approach to Computer Networks-S.Keshav, 2nd Ed, Pearson Education
2. Understanding communications and Networks, 3rd Edition, W.A. Shay, Thomson

Targeted Proficiency Level (For each course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5	CO6
Targeted Proficiency Level		60	60	60	60	60	60
Targeted level of Attainment	Level 3	60	60	60	60	60	60
	Level 2	50	50	50	50	50	50
	Level 1	40	40	40	40	40	40

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 1	Dissemination of Vision, Mission of the Dept. and PEOs, POs,& PSOs of the Programme		1	Lecture	ICT
2		Describe Reference models- The OSI Reference Model	K1	2	Lecture	ICT
3		Discuss TCP/IP Reference Model	K2	1	Lecture with Discussion	ICT
4		Examples of Networks: ARPANET, INTERNET	K2	1	Lecture	ICT
5		Describe Network Topologies WAN, LAN, MAN.	K1	1	Lecture	ICT

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 2	Describe Physical Layer – Fourier Analysis, The Maximum Data	K1	1	Lecture	ICT
2		Explain Different Kinds of Transmission Media	K2	2	Lecture	ICT
3		Describe Multiplexing Techniques: Frequency Division Multiplexing and Wavelength Division Multiplexing	K1	2	Lecture	ICT
4		Discuss Time Division Multiplexing	K1	1	Lecture with Discussion	ICT
5		Explain Circuit Switched Networks	K2	1	Lecture with Discussion	ICT
6		Explain Datagram Networks and Virtual Circuit Networks	K2	2	Lecture with Discussion in class Assignment	ICT

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	Describe The Data Link Layer - Services Provided to the Network Layer	K1	1	Lecture	ICT
2		Explain Data Link Layer Design Issues	K2	1	Lecture	ICT

3		Illustrate Framing – Error Control –Flow Control, Error Detection and Correction – Error-Correcting Codes – Error Detecting code	K3	2	Lecture with Discussion and in class Assignment	ICT
4		Discuss MAC Protocols: ALOHA, CSMA	K2	2	Lecture with Discussion	ICT
5		Explain Elementary Data Link Protocols- A Utopian Simplex Protocol	K2	1	Lecture with Discussion	ICT
6		Discuss A Simplex Stop and Wait Protocol for an Error free channel-A Simplex Stop and Wait Protocol for a Noisy Channel.	K2	1	Lecture	ICT
7		Demonstrate Sliding Window Protocols	K3	3	Lecture with Discussion	ICT
8		Discuss HDLC, PPP and Piggybacking	K2	2	Lecture	ICT

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 4	Describe The Network Layer Design Issues Store-and-Forward Switching, Services Provided to Transport Layer	K1	2	Lecture	ICT
2		Discuss Implementation of Connectionless Service-Implementation of Connection Oriented Service.	K2	2	Lecture with Discussion	ICT
3		Construct Shortest Path algorithm	K3	1	Lecture with Discussion	ICT
4		Describe Flooding	K2	1	Lecture	ICT
5		Illustrate Distance Vector Routing and Hierarchical Routing	K3	2	Lecture and in class Assignment	ICT
6		Differentiate Broadcast and Multicast Routing	K2	1	Lecture	ICT
7		Discuss Congestion Control Algorithms	K2	1	Lecture with Discussion	ICT
8		Demonstrate IP Addressing and Subnet Masking	K3	1	Lecture with Discussion and in class Assignment	ICT

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 5	Describe Transport Layer – Services, Primitives and Sockets	K1	2	Lecture	ICT
2		Discuss Elements of Transport Protocols	K2	2	Lecture	ICT
3		Discuss The Internet Transport Protocols: TCP Segment Header and Primitives	K2	2	Lecture with Discussion	ICT
4		Discuss The Internet Transport Protocols: UDP, RPC, RTP, and RTCP Segment Headers and Primitives	K2	1	Lecture with Discussion	ICT
5		Discuss Congestion Control in TCP	K2	1	Lecture with Discussion	ICT
6		Outline Quality of Service	K1	1	Lecture	ICT
7		Discuss QoS Improving Techniques: Leaky Bucket and Token Bucket Algorithms	K2	1	Lecture with Discussion	ICT

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 6	Discuss Application Layer – DNS: The DNS Name Space, Resource Records, Name Servers.	K2	1	Lecture with Discussion	ICT
2		Describe Electronic Mail Message Transfer Agent: SMTP and Message Access Agent: POP	K2	2	Lecture	ICT
3		Discuss FTP: Control Connection and Data Connection	K2	1	Lecture with Discussion	ICT
4		Discuss HTTP: Connections, Methods, Message Headers, and Caching	K2	1	Lecture with Discussion	ICT
5		Describe Cryptography and Authentication	K2	1	Lecture	ICT
6		Differentiate Public Key and Private Key Cryptographic	K2	2	Lecture with Discussion	ICT

		Algorithms				
7		Discuss Digital Signatures and Certificates	K2	1	Lecture with Discussion	ICT
8		Outline the working of Firewalls	K2	1	Lecture	ICT
		Discuss Wireless Security	K2	1	Lecture	ICT

Total No. of Classes: 60

Artificial Intelligence

Academic Year: 2021-22

Year/ Semester: V

Name of the Course: Artificial Intelligence

Programme: B.Tech

Section: A,B,C& D

Course Code: V18CST17

LESSON PLAN

Course Outcomes (Along with Knowledge Level): After completion of this course, Student will be able to:

CO	Course Outcomes	Knowledge Level
1	Illustrate the concept of Intelligent systems and current trends in AI	K2
2	Apply Problem solving, Problem reduction and Game Playing techniques in AI	K3
3	Illustrate the Logic concepts in AI	K2
4	Explain the Knowledge representation techniques in AI	K2
5	Describe Expert systems and their applications	K2
6	Illustrate Uncertainty Measures	K2

1. Artificial Intelligence, SarojKaushik, 1st Edition, Cengage Learning

REFERENCE BOOKS:

1. Artificial Intelligence, Elaine Rich, Kevin Knight, Shivashankar B Nair, 3rd Edition, Tata McGraw Hill Education Private Limited., 2009
2. Artificial Intelligence- A modern Approach, 3rd Edition, Stuart Russel, Peter Norvig, Pearson Education

Targeted Proficiency and Attainment Levels (for each course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5	CO6
Targeted Proficiency Level		60	60	60	60	60	60
Targeted level of Attainment	Level 3	60	60	60	60	60	60
	Level 2	50	50	50	50	50	50
	Level 1	40	40	40	40	40	40

Lecture Plan:

S.No	Course Outcome	Intended Learning Outcomes(ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 1	Dissemination of Department Vision, Mission, PEOs, POs, PSOs	-	-	-	
		Explain AI Introduction and history	K1	1	Lecture	ICT
2		Describe Intelligent Systems	K2	1	Lecture With Discussion	ICT
3		Discuss foundations of AI	K2	1	Lecture With Discussion	ICT
4		Describe AI applications	K2	1	Lecture With Discussion	ICT
5		Illustrate Tic-Tac-Toe game playing approaches	K2	2	Lecture With Discussion	ICT
6		Describe current trends in AI	K2	1	Lecture With Discussion	ICT

07

S.No	Course Outcome	Intended Learning Outcomes(ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 2	Illustrate general problem solving approaches and problem characteristics	K3	2	Lecture with Discussion and in class Assignment	ICT
2		Demonstrate exhaustive searches	K3	2	Lecture with Discussion and in class Assignment	ICT
3		Demonstrate Heuristic search techniques	K3	3	Lecture with Discussion and in class Assignment	ICT
4		Illustrate Iterative Deepening A* and constraint satisfaction	K3	2	Lecture with Discussion and in class Assignment	ICT
5		Demonstrate problem reduction approaches	K3	1	Lecture with Discussion and in class Assignment	ICT
6		Demonstrate Game playing approaches	K3	2	Lecture with Discussion and in class Assignment	ICT
7		Use Bounded look-ahead strategy, evaluation functions and MINIMAX procedure	K3	1	Lecture with Discussion and in class Assignment	ICT
8		Illustrate Alpha-Beta pruning and two player	K3	1	Lecture with Discussion and in class	ICT

		perfect information games			Assignment	
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14

S.No	Course Outcome	Intended Learning Outcomes(ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	Explain Propositional calculus	K2	1	Lecture with Discussion	BB/ICT
2		Demonstrate Propositional Logic	K2	1	Lecture with Discussion	BB/ICT
3		Demonstrate Natural Deduction system	K2	1	Lecture with Discussion	BB/ICT
4		Explain Axiomatic system	K2	1	Lecture with Discussion	BB/ICT
5		Demonstrate semantic tableau system in propositional logic	K2	1	Lecture with Discussion	BB/ICT
6		Explain resolution refutation in propositional logic	K2	2	Lecture with Discussion	BB/ICT
7		Demonstrate predicate logic	K2	3	Lecture with Discussion	BB/ICT

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S.No	Course Outcome	Intended Learning Outcomes(ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 4	Explain approaches to knowledge representation	K2	1	Lecture with Discussion	BB/ICT
2		Demonstrate Knowledge Representation(KR) using semantic network	K2	2	Lecture with Discussion	BB/ICT
3		Demonstrate Extended semantic networks for KR	K2	2	Lecture with Discussion	BB/ICT
4		Describe Knowledge Representation using Frames	K2	2	Lecture with Discussion	BB/ICT

07

S.No	Course Outcome	Intended Learning Outcomes(ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 5	Discuss phases in building expert systems	K2	2	Lecture with Discussion	BB/ICT
2		Explain Expert systems versus Traditional systems	K2	2	Lecture with Discussion	BB/ICT
3		Discuss Rule-based Expert systems	K2	2	Lecture with	BB/ICT

					Discussion	
4		Explain Blackboard systems	K2	2	Lecture with Discussion	BB/ICT
5		Explain Truth Maintenance Systems	K2	2	Lecture with Discussion	BB/ICT
6		List applications of expert systems	K2	1	Lecture with Discussion	BB/ICT

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S.No	Course Outcome	Intended Learning Outcomes(ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 6	Explain Probability Theory	K2	3	Lecture with Discussion	BB/ICT
2		Explain Bayesian Belief Networks	K2	2	Lecture with Discussion	BB/ICT
3		Discuss Certainty Factor theory	K2	1	Lecture with Discussion	BB/ICT
4		Discuss Dempster-Shafer theory	K2	2	Lecture with Discussion	BB/ICT

08

Total No. of Classes: 57

Design and Analysis of Algorithms

Academic Year: 2020-21

Programme: B.Tech

Year/ Semester: V

Section: A,B,C& D, (CST)

Name of the Course: Design and Analysis of Algorithms Course Code:V18CST13/304

LESSON PLAN

Course Outcomes (Along with Knowledge Level): After completion of this course, Student will be able to:

S.No.	CO No.	Course Outcome	BTL
1.	304.1	Describe asymptotic notations and basic concepts of algorithms.	K2
2.	304.2	Use divide and conquer paradigm to solve various problems.	K3
3.	304.3	Use greedy technique to solve various problems.	K3
4.	304.4	Demonstrate dynamic programming technique to various problems.	K3
5.	304.5	Employ backtracking technique to various problems.	K3
6.	304.6	Demonstrate branch and bound technique to various problems.	K3

Textbooks:

- Fundamentals of Computer Algorithms - Ellis Horowitz, Satraj Sahni and Rajasekaran; 2/e; Universities Press.
- Design and Analysis of Algorithms - S Sridhar; Oxford.

Reference Books:

- Design and Analysis of Algorithms - Aho, Ullman and Hopcroft; Pearson Education.
- Introduction to Algorithms - T.H.Cormen, C.E.Leiserson, R.L.Rivest and C. Stein; 2/e; PHI

Targeted Proficiency and attainment Levels (for each Course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5	CO6
Targeted Proficiency Level(Marks In %)		60	60	60	60	60	60
Targeted level of Attainment	Level 3	75	80	70	70	75	70
	Level 2	70	75	65	65	70	65
	Level 1	65	70	60	60	65	60

Lecture Plan:

UNIT - 1: Introduction						
S. No.	Course Outcome	Intended Learning Outcome (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching Aid
1	V18CST13.1	Dissemination of Department Vision, Mission PO's, PSO's and CO's.		01	Lecture	ICT/BB
2		Define Algorithm,	K1	01	Lecture	ICT/BB

		Properties of Algorithm				
3		Discuss Algorithm Specification-Pseudo Code Conventions, Recursive Algorithms	K2	01	Lecture	ICT/BB
4		Explain Performance Analysis-Space Complexity, Time Complexity	K2	02	Lecture	ICT/BB
5		Describe Asymptotic Notations- Big oh notation, Omega notation, Theta notation and Little oh notation	K2	01	Lecture	ICT/BB
6		Estimate Practical Complexities	K2	02	Lecture	ICT/BB
7		Explain Amortized Analysis, Probabilistic Analysis	K2	01	Lecture	ICT/BB
8		Estimate Performance Measurement	K2	01	Lecture	ICT/BB
			Total:	10		

UNIT - 2: Divide and Conquer						
S. No.	Course Outcome	Intended Learning Outcome (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching Aid
1	V18CST13.2	Describe Divide and Conquer strategy, Control Abstraction	K1	01	Lecture	ICT/BB
2		Use D&C to Find the minimum and maximum	K2	01	Lecture	ICT/BB
3		Use Divide and Conquer to Binary Search	K3	01	Lecture + Discussion	ICT/BB
4		Apply Divide and Conquer to Quick Sort	K3	02	Lecture + Discussion	ICT/BB
5		Apply Divide and Conquer to Merge Sort, Refinements to merge sort	K3	02	Lecture + Discussion	ICT/BB
			Total:	07		

UNIT - 3: Greedy Method						
S. No.	Course Outcome	Intended Learning Outcome (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching Aid
1	V18CST13.3	Describe Greedy method, Control Abstraction	K1	01	Lecture	ICT/BB

2		Solve Job Sequencing with deadlines	K3	02	Lecture + Discussion	ICT/BB
3		Solve Knapsack Problem	K3	02	Lecture + Discussion	ICT/BB
4		Explain Spanning Trees, Find Minimum Cost Spanning Trees(Prim's and Kruskal's algorithms)	K3	02	Lecture + Discussion	ICT/BB
5		Solve Single Source Shortest Path Problem	K3	01	Lecture + Discussion	ICT/BB
6		Choose Optimal Merge Patterns	K3	01	Lecture + Discussion	ICT/BB
			Total:	09		

UNIT - 4: Dynamic Programming						
S. No.	Course Outcome	Intended Learning Outcome (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching Aid
1	V18CST13.4	Describe Dynamic Programming	K1	01	Lecture	ICT/BB
2		Solve All Pairs Shortest Path Problem	K3	01	Lecture + Discussion	ICT/BB
3		Solve Single Source shortest Paths General Weights	K3	01	Lecture + Discussion	ICT/BB
4		Explain Optimal Binary Search Trees	K3	03	Lecture + Discussion	ICT/BB
5		Explain String Edition	K3	02	Lecture + Discussion	ICT/BB
6		Solve 0/1 Knapsack Problem	K3	01	Lecture + Discussion	ICT/BB
7		Solve Reliability Design	K3	03	Lecture + Discussion	ICT/BB
			Total:	12		

UNIT - 5: Backtracking						
S. No.	Course Outcome	Intended Learning Outcome (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching Aid
1	V18CST13.5	Explain General method	K2	01	Lecture	ICT/BB
2		Solve N-Queen Problem	K3	02	Lecture + Discussion	ICT/BB
3		Employ Backtracking to solve Sum of Subsets Problem	K3	03	Lecture + Discussion	ICT/BB

4		Demonstrate Graph Colouring	K3	02	Lecture + Discussion	ICT/BB
5		Demonstrate Hamiltonian Cycles	K3	02	Lecture + Discussion	ICT/BB
			Total:	10		

UNIT - 6: Branch and Bound						
S. No.	Course Outcome	Intended Learning Outcome (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching Aid
1	V18CST13.6	Describe the General method of Branch and Bound technique	K1	01	Lecture	ICT/BB
2		Solve Travelling Salesperson problem	K3	02	Lecture + Discussion	ICT/BB
3		Apply Branch and Bound to 0/1 Knapsack problem	K3	03	Lecture + Discussion	ICT/BB
4		Demonstrate LC Branch and Bound solution	K3	03	Lecture + Discussion	ICT/BB
5		Demonstrate the FIFO Branch and Bound solution	K3	03	Lecture + Discussion	ICT/BB
			Total:	12		

Total Number of Hours: 60

Database Management Systems

Academic Year: 2020-21

Year/ Semester: V

Name of the Course: Database Management Systems

Programme: B.Tech

Section: A,B,C,D& CST

Course Code:V18CST10

LESSON PLAN

Course Outcomes (Along with Knowledge Level):After completion of this course, students will be able to:

S.No.	CO No.	Course Outcome	BTL
1.	V18CST10.1	Demonstrate Database Systems, various Data Models and Database Architecture.	K2
2.	V18CST10.2	Apply ER Modelling to Design Relational Databases for Real Time Applications.	K3
3.	V18CST10.3	Apply SQL Constructs to Perform Database Operations.	K3
4.	V18CST10.4	Apply Normalization Techniques to Refine Schema.	K3
5.	V18CST10.5	Explain Transaction Management and Concurrency Control.	K2
6.	V18CST10.6	Experiment with various database indexing techniques.	K3

Text Books:

1. Introduction to Database Systems, CJ Date, 8th Edition, Pearson Education.
2. Database Management Systems, Raghurama Krishnan, Johannes Gehrke, 3rd Edition TATA McGraw Hill.

Reference Books:

1. Database Systems design, Implementation, and Management, Peter Rob & Carlos Coronel 7th Edition, Course Technology.
2. Fundamentals of Database Systems, ElmasriNavrate, 7th Edition, Pearson Education.
3. Database Systems - The Complete Book, H G Molina, J D Ullman, J Widom, 2nd Edition, Pearson

Targeted Proficiency and attainment Levels (for each Course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5	CO6
Targeted Proficiency Level		65	65	60	60	60	65
Targeted level of Attainment	Level 3	65	60	60	60	60	65
	Level 2	60	55	55	55	55	60
	Level 1	55	50	50	50	50	55

Lecture Plan:

UNIT-I: AN OVERVIEW OF DATABASE SYSTEMS

SN0	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1	CO1:	Dissemination of vision, mission, PEOs, POs, PSOs		1	Lecture	BB
2		Define a Database System, Database	K1	1	Lecture	BB
3		Describe the disadvantages in Traditional File System and advantages of DBMS over file system	K1	1	Lecture with Discussion	BB+ICT
4		Describe data independence	K1	1	Lecture with Discussion	BB
5		Discuss Database Architecture: The Three levels of the architecture: The External level, Conceptual level, Internal level and Mappings	K2	2	Lecture with Discussion	BB+ICT
6		Illustrate Database Management System Architecture and components	K2	1	Lecture with Discussion	BB
7		Explain different types of users in DBMS	K2	1	Lecture with Discussion	BB
8		Illustrate the structure of DBMS	K2	1	Lecture with Discussion	BB+ICT
9		Explain Client/Server Architecture	K2	1	Lecture with Discussion	BB
		TOTAL		10		

UNIT-II: DATABASE DESIGN, RELATIONAL MODEL, RELATIONAL ALGEBRA AND CALCULUS

SN0	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1	CO2:	Describe Introduction to Database Design: Database Design and ER Diagrams	K1	1	Lecture	BB
2		Explain Entities, Attributes and Entity Sets	K2	1	Lecture With Discussion	BB
3		Explain Relationships and Relationship Sets	K2	1	Lecture With Discussion	BB
4		Describe Relational Systems and Others	K2	2	Lecture With Discussion	BB +IC T
5		Illustrate Conceptual Design with the ER Model	K3	1	Lecture With Discussion	BB +IC T
6		Explain The Relational Model: Integrity Constraints over Relations: Key Constraints, Foreign Key Constraints, General Constraints	K2	2	Lecture With Discussion	BB +IC T
7		Illustrate Relational Algebra: Selection and Projection	K3	1	Lecture With Discussion	BB
8		Illustrate Set Operations, Renaming, Joins, Division, More Example of Algebra Queries	K3	3	Lecture With Discussion	BB +IC T
9		Illustrate Relational Calculus: Tuple Relational Calculus, Domain Relational Calculus	K3	1	Lecture With Discussion	BB
		TOTAL		13		

UNIT-III: QUERIES, CONSTRAINTS, TRIGGERS						
S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	Explain the Form of a Basic SQL Query	K2	2	Lecture with Demonstration	BB+ICT
2		Illustrate Union, Intersect, Except and Nested Queries	K3	3	Lecture with Demonstration	BB+ICT
3		Illustrate Aggregate Operators	K3	2	Lecture with Demonstration	BB
4		Interpret Null Values	K2	1	Lecture with Demonstration	BB
5		Illustrate Complex Integrity Constraints in SQL	K3	1	Lecture with Demonstration	BB
6		Illustrate Triggers and Active Databases	K3	2	Lecture with Demonstration	BB+ICT
		TOTAL		11		

UNIT-IV: SCHEMA REFINEMENT (NORMALIZATION)						
S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 4	Explain Purpose of Normalization or Schema Refinement	K2	1	Lecture With Discussion	BB
2		Explain the Concept of Functional Dependency	K2	2	Lecture With Discussion	BB+ICT
3		Illustrate Normal Forms Based on Functional Dependency (1NF, 2NF and 3	K3	3	Lecture With Discussion	BB+I

		NF)				CT
4		Demonstrate Concept of Surrogate Key, Boyce-Codd Normal Form (BCNF)	K3	1	Lecture With Discussion	BB+I CT
5		Illustrate Lossless Join and Dependency Preserving Decomposition, Fourth Normal Form (4NF)	K3	2	Lecture With Discussion	BB+I CT
		TOTAL		9		

UNIT-V: TRANSACTION MANAGEMENT AND CONCURRENCY CONTROL

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 5	Describe Transaction, Properties of Transactions, Transaction Log	K2	1	Lecture With Discussion	BB
2		Explain Transaction Management with SQL Using Commit, Rollback and Save point.	K2	2	Lecture With Discussion	BB+ICT
3		Describe Concurrency Control for Lost Updates, Uncommitted Data, Inconsistent Retrievals and The Scheduler	K2	2	Lecture With Discussion	BB+ICT
4		Explain Concurrency Control with Locking Methods: Lock Granularity, Lock Types, Two Phase Locking for Ensuring Serializability	K2	3	Lecture With Discussion & Seminar	BB+ICT
5		Explain Deadlocks, Concurrency Control with Time Stamp Ordering	K2	2	Lecture With Discussion	BB+ICT
6		Explain Wait/Die and Wound/Wait Schemes	K2	1	Lecture With Discussion	BB
7		Describe Database Recovery Management: Transaction	K2	1	Lecture With Discussion	BB

		Recovery				
		TOTAL		12		

UNIT-VI: STORAGE AND INDEXING

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1		Describe Overview of Storages and Indexing	K1	1	Lecture With Discussion	BB+ICT
2	CO 6	Describe data on External Storage	K1	1	Lecture With Discussion	BB+ICT
3		Explain File Organizations and Indexing: Clustered Indexes, Primary and Secondary Indexes	K2	2	Lecture With Discussion	BB
4		Explain Index Data Structures: Hash-Based Indexing, Tree-Based Indexing	K2	2	Lecture With Discussion	BB
5		Explain Comparison of File Organizations	K2	2	Lecture With Discussion	BB+ICT
		TOTAL		8		

Total No. of Classes: 63

Organizational Behaviour

Academic Year: 2021-22

Year/ Semester: V

Name of the Course: Organizational Behaviour

Programme: B.Tech

Section: A,B,C& D

Course Code: V18MBT53

LESSON PLAN

COURSE OUTCOMES (Along with Knowledge Level):

After completion of this course, the students will be able to:

S.No.	CO No.	Course Outcome	BTL
1.	CO 1	Identify the basic concepts of organizational behaviour	K1
2.	CO 2	Enumerate the importance of individual personality and learning in the organizational context.	K2
3.	CO 3	Develop decision making abilities and interpersonal communication skills	K3
4.	CO 4	Identify the basic concepts of Group dynamics	K1
5.	CO 5	Identify the foundations of organization development	K1
6.	CO 6	Develop team building skills	K2

Text Books:

1. Organizational behaviour, Stephen P Robbins, Timothy A. Judge, Neharika Vohra, 14th Edition, Pearson, 2012

Reference Books:

1. K.Aswathappa: "Organizational Behavior-Text, Cases and Games", Himalaya Publishing House, New Delhi, 2008,
2. Steven L McShane, Mary Ann Von Glinow, Radha R Sharma: "Organizational Behavior", Tata McGraw Hill Education, New Delhi, 2008.

3. Targeted Proficiency and attainment Levels (for each Course Outcome):

CO'S		CO1	CO2	CO3	CO4	CO5	CO6
Targeted Proficiency Level		60	60	60	60	60	60
Targeted level of Attainment	Level 3	70	70	70	70	70	70
	Level 2	60	60	60	60	60	60
	Level 1	50	50	50	50	50	50

*Figures indicated above represents percentile

Lecture Plan

Unit: I

S#	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1	CO1: K1	Define OB	K1	1	Lecture	BB
2		Explain the Nature and scope	K2	2	Lecture + discussion	BB
3		Explain how OB linked with other social sciences	K2	2	Lecture	BB
4		Explain the concept of Individual Roles & Organizational Goals	K2	2	Lecture + discussion	BB
5		Describe Perspectives of Human Behaviour	K2	1	Lecture	BB
6		Describe various approaches to organizational behaviour	K2	2	Lecture	BB
7		Explain models of organizational behaviour	K2	2	Lecture	BB
Total				12		

UNIT- II

1	CO2:K2	Define Personality	K1	1	Lecture	BB
2		Explain Nature, stages and determinants of personality	K2	1	Lecture + discussion	BB
3		Explain Johari Window	K2	1	Lecture+Discussion	BB
4		Explain Transaction Analysis	K2	1	Lecture + discussion	BB
5		Describe	K2	1	Lecture	BB

		Learning process				
6		Explain Learning Theories	K2	1`	Lecture	BB
7		Explain Perception nature and process	K2	1	Lecture	BB
8		Explain Motivation concepts and theories	K2	1	Lecture	BB
9		Describe Attitude and values	K2	1	Lecture	BB
Total				9		

Unit - III

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Reqd.	Pedagogy	Teaching aids
1.	CO3: K3	Define Decision Making	K1	1	Lecture	BB
2.		Explain Decision Making Process	K2	1	Lecture + discussion	BB
3.		Explain Individual Decision Making	K2	1	Lecture + discussion	BB
4.		Explain Group Decision Making	K2	1	Lecture + discussion	BB
5.		Define interpersonal communication	K1	2	Lecture + discussion	BB
6.		Explain Types of Decision Making	K2	1	Lecture + discussion	BB
		Total		7		

UNIT - IV

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Reqd.	Pedagogy	Teaching aids
1.	CO4:K1	Define Group Dynamics	K1	1	Lecture+ discussion	BB
2.		Describe behavioural dimensions	K2	1	Lecture + discussion and	BB+P PT
3.		Explain Groups and their formation	K2	2	Lecture + discussion	BB
4.		Explain Informal Organizations,	K2	1	Lecture	BB
5		Explain Group versus Individual Interaction	K2	1`	Lecture	BB

6		Describe Listening, Feedback, Collaborative Processes in Work Groups	K2	2	Lecture	BB
7		Explain Conflict Resolution in Groups and Problem Solving Techniques.	K2	2	Lecture	BB
		Total		10		

UNIT - V

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Reqd.	Pedagogy	Teaching aids
1.	C05:K1	Define Goals, processes	K1	1	Lecture + discussion	BB
2.		Describe change	K2	2	Lecture + discussion	BB
3.		Explain resistance to change	K2	2		
4.		Explain Impact of technology on human behaviour	K2	1	Lecture	BB
5.		Explain Nature of OD - interventions	K2	1	Lecture + discussion and In-class Assignment	BB+PPT
6		Explain OD techniques and OD applications.	K2	2	Lecture	BB
		Total		09		

UNIT - VI

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Reqd.	Pedagogy	Teaching aids
1.	C06:K2	Define Team Building	K1	1	Lecture + discussion	BB
2.		Explain The Significance of Team Building	K2	2	Lecture + discussion	BB
3.		Explain Stages of Team Building	K2	2	Lecture + discussion	BB+PPT
4.		List the benefits of Team Building	K1	1	Lecture	BB
5.		Discuss High Performance Teams	K2	1	Lecture + discussion	BB
		Total		7		

Operating Systems

Academic Year: 2021-22

Year/ Semester: V

Name of the Course: Operating Systems Code: V18CST12

Programme: B.Tech

Section: A,B,C& D

LESSON PLAN

Course Outcomes: Upon Completion of the course, student will be able to:

S.No.	Co No.	Course Outcome	Knowledge Level
1.	CO1	Describe Operating System Services and System Calls	K2
2.	CO2	Illustrate Process Management Concepts and CPU Scheduling Algorithms	K3
3.	CO3	Demonstrate Process Synchronization primitives	K3
4.	CO4	Demonstrate Deadlock Prevention, Avoidance and Detection methods	K3
5.	CO5	Illustrate Memory Management Techniques and Page Replacement Algorithms	K3
6.	CO6	Describe File System Concepts and Mass Storage Structures	K2

Text Books:

1. Operating System Concepts, AbrahamSilberschatz, ,Peter Baer Galvin,Greg Gagne, 9th Edition, John Wiley and Sons Inc., 2012

Reference Books:

1. Operating Systems – Internals and Design Principles, William Stallings, 7th Edition, Prentice Hall, 2012
2. Modern Operating Systems, Andrew S. Tanenbaum, Third Edition, Addison Wesley,2007

Targeted Proficiency and attainment Levels (for each Course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5	CO6
Targeted Proficiency Level		65	65	65	60	60	60
Targeted level of Attainment	Level 3	65	65	65	60	60	60
	Level 2	55	55	55	50	50	50
	Level 1	45	45	45	40	40	40

Lecture Plan:

SNO	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours required	Pedagogy	Teaching aids
1	CO 1	Dissemination of Department Vision, Mission, PEOs, POs, PSOs	-	-	-	
		Introduction of OS	K2	1	Lecture with Discussion	BB/ICT
2		Operating-System Structure	K2	1	Lecture with Discussion	BB/ICT
3		Operating-System Services	K2	2	Lecture with Discussion	BB/ICT
4		User and Operating-System Interface	K2	1	Lecture with Discussion	BB/ICT
5		System Calls	K2	1	Lecture with Discussion	BB/ICT
6		Types of System Calls	K2	1	Lecture with Discussion	BB/ICT

SNO	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours required	Pedagogy	Teaching aids
1	CO2	Process Scheduling	K3	2	Lecture with Discussion and in class Assignment	ICT
2		Operations On Processes	K3	1	Lecture with Discussion and in class Assignment	ICT
3		InterProcessCommunication(IPC)	K3	1	Lecture with Discussion and in	ICT

					class Assignmen t	
4		Threads Introduction	K3	1	Lecture with Discussion and in class Assignmen t	ICT
5		Multithreading Models	K3	1	Lecture with Discussion and in class Assignmen t	ICT
6		CPU Scheduling Basic Concepts	K3	1	Lecture with Discussion and in class Assignmen t	ICT
7		CPU Scheduling Criteria	K3	1	Lecture with Discussion and in class Assignmen t	ICT
8		CPU Scheduling Algorithms	K3	4	Lecture with Discussion and in class Assignmen t	ICT

SN O	Course Outcome	Intended Learning Outcomes (ILO)	Knowledg e Level of ILO	No. of Hours required	Pedagogy	Teach ing aids
1	C03	Critical Section Problem	K3	1	Lecture with Discussion and in class Assignment	ICT
2		Peterson's Solution			Lecture with Discussion	ICT

			K3	1	and in class Assignment	
3		Synchronization Hardware	K3	1	Lecture with Discussion and in class Assignment	ICT
4		Mutex Locks	K3	2	Lecture with Discussion and in class Assignment	ICT
5		Semaphores	K3	1	Lecture with Discussion and in class Assignment	ICT
6		Classic Problems of Synchronization	K3	2	Lecture with Discussion and in class Assignment	ICT
7		Monitors	K3	2	Lecture with Discussion and in class Assignment	ICT

SNO	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours required	Pedagogy	Teaching aids
1	C04	Dead Lock Introduction	K3	1	Lecture with Discussion and in class Assignment	ICT
2		System Model	K3	1	Lecture with Discussion and in class Assignment	ICT
3		Deadlock Characterization	K3	1	Lecture with Discussion and in class Assignment	ICT
4		Methods for Handling Deadlocks,	K3	2	Lecture with Discussion and in class Assignment	ICT
5		Deadlock Prevention	K3	2	Lecture with Discussion and in class Assignment	ICT
6		Deadlock Avoidance		1	Lecture with Discussion	ICT

			K3		and in class Assignment	
7		Deadlock Detection	K3	2	Lecture with Discussion and in class Assignment	ICT
8		Recovery from Deadlock	K3	2	Lecture with Discussion and in class Assignment	ICT

SN O	Course Outcome	Intended Learning Outcomes (ILO)	Knowledg e Level of ILO	No. of Hours required	Pedagogy	Teach ing aids
1	C05	Swapping	K3	1	Lecture with Discussion and in class Assignment	ICT
2		Contiguous Memory Allocation	K3	1	Lecture with Discussion and in class Assignment	ICT
3		Segmentation,	K3	1	Lecture with Discussion and in class Assignment	ICT
4		Paging,	K3	1	Lecture with Discussion and in class Assignment	ICT
5		Structure of the Page Table	K3	1	Lecture with Discussion and in class Assignment	ICT
6		Demand Paging	K3	1	Lecture with Discussion and in class Assignment	ICT
7		Page Replacement	K3	3	Lecture with Discussion and in class Assignment	ICT
8		Allocation of Frames	K3	1	Lecture with Discussion and in class Assignment	ICT

9		Thrashing	K3	1	Lecture with Discussion and in class Assignment	ICT
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SN O	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours required	Pedagogy	Teaching aids
1	C06	Overview of Mass-Storage Structure	K2	1	Lecture with Discussion	BB/ICT
2		Disk Scheduling	K2	2	Lecture with Discussion	BB/ICT
3		File Concept, Access Methods	K2	1	Lecture with Discussion	BB/ICT
4		Directory and Disk Structure	K2	1	Lecture with Discussion	BB/ICT
5		File-System Mounting	K2	1	Lecture with Discussion	BB/ICT
6		File Allocation Methods	K2	1	Lecture with Discussion	BB/ICT

Unix Programming

Academic Year: 2021-22

Year/ Semester: V

Name of the Course: Unix Programming

Programme: B.Tech

Section: A,B,C& D

Course Code: V18CST14/ C-305

LESSON PLAN

Course Outcomes (Along with Knowledge Level):

After completion of this course, students will be able to:

CO	Course Outcomes	Knowledge Level
1	Illustrate the UNIX basics and the working of the built in commands in Unix	K2
2	Demonstrate the file system and change the permissions associated with files	K2
3	Develop basic programs using shell script	K3
4	Demonstrate the grep family and data transforming programs sed, and awk	K2
5	Construct programs for process system calls	K3
6	Explain the concept of signals and its system call	K2

Text Books:

1. Introduction to Unix and shell programming, M G venkateshmurthy, Pearson education
2. Advanced programming in the unix environment, W. Richard Stevens, 3rd Edition, Pearson education.

REFERENCES

1. Unix and shell Programming, B.A. Forouzan& R.F. Giberg, Thomson, First Edition, NewDelhi, 2003.

Targeted Proficiency and Attainment Levels (for each course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5	CO6
Targeted Proficiency Level		60	60	60	60	60	60
Targeted level of Attainment	Level 3	60	60	60	60	60	60
	Level 2	50	50	50	50	50	50
	Level 1	40	40	40	40	40	40

Lecture Plan:

UNIT - I

S#	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1.	CO1	Dissemination of	K1	1	Lecture	PPT

		Department Vision, Mission, PEOs, POs, PSOs, COs, Introduction to Unix Programming and Describe Brief History			with discussion	
2.		Explain The UNIX Architecture and Basic features of UNIX	K2	1	Lecture with discussion	PPT
3.		Demonstrate General Purpose Utilities: cal, date, man, echo, bc, clear, passwd, who, whoami, uname and Directory Handling commands: pwd, cd, mkdir, rmdir	K2	1	Lecture with demonstration	PPT, CLI
4.		Demonstrate File Handling Utilities: cat, touch, cp, ls, rm, mv, nl, pg, tar, wc	K2	2	Lecture with demonstration	PPT, CLI
5.		Demonstrate Displaying Commands: more, head, tail and simple filter commands: cmp, comm., ulink, diff, head, tail, find, cut, paste, sort, uniq, tr, finger	K2	2	Lecture with demonstration	PPT, CLI
6.		Demonstrate Disk Utilities: du, df, mount, umount and Process Utilities: ps, kill	K2	1	Lecture with demonstration	PPT, CLI
7.		Demonstrate Networking Utilities- ping, telnet, rlogin, ftp	K2	1	Lecture with demonstration	PPT, CLI
Total Hours				9 Hours		

UNIT – II

S#	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1.	CO2	Discuss the Types of Files, Directories and Files,	K2	1	Lecture with discussion	PPT
2.		Explain the UNIX File	K2	1	Lecture	PPT

		System and Absolute and relative pathnames			with discussion	
3.		Discuss the File Attributes and Permissions	K2	1	Lecture with discussion	PPT
4.		Discuss File Command knowing the File Type	K2	1	Lecture with discussion	PPT
5.		Demonstrate Chmod Command Changing File Permissions	K2	2	Lecture with demonstration	PPT, CLI
6.		Demonstrate Chown Command Changing the Owner of a File and Chgrp Command Changing the Group of a File.	K2	1	Lecture with demonstration	PPT, CLI
7.		Explain Vi editor- editing with vi, moving the cursor, editing, copying and moving text, pattern searching.	K2	1	Lecture with demonstration	PPT, Live Demo
Total Hours				8 Hours		

UNIT - III

S#	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1.	CO3	Explain Shell Variables	K2	2	Lecture with discussion	PPT
2.		Examine The Export Command and Explain The .Profile File a Script Run During Starting	K2	1	Lecture with discussion	PPT
4.		Demonstrate The First Shell Script and read Command	K3	1	Lecture with demonstration	PPT
5.		Explain the Positional parameters and Explain The \$?Variable knowing the exit Status	K2	2	Lecture with discussion	PPT

6.		Demonstrate More about the Set Command and Practice The Exit Command	K3	1	Lecture with demonstration	PPT
7.		Practice on Branching Control Structures	K3	3	Lecture with demonstration	PPT
8.		Practice on Loop Control Structures	K3	2	Lecture with demonstration	PPT
9.		Examine The Continue and Break Statement	K3	1	Lecture with demonstration	PPT
10.		Demonstrate Expr Command: Performing Integer Arithmetic and Real Arithmetic in Shell programs	K3	2	Lecture with demonstration	PPT
11.		Explain The here document(<<) and Demonstrate the Sleep Command	K2	1	Lecture with demonstration	PPT
12.		Demonstrate Debugging Scripts and Demonstrate the Eval Command	K3	1	Lecture with demonstration	PPT
13.		Explain The ExecCommand.	K2	1	Lecture with demonstration	PPT
14.		Illustrate Command Line Structure	K3	1	Lecture with discussion	PPT
15.		Demonstrate Meta characters	K3	1	Lecture with demonstration	PPT
Total Hours				20 Hours		

UNIT - IV

S#	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1.	CO4	Demonstrate the Grep Family-egrep, fgrep	K2	2	Lecture with	PPT, CLI

					demonstration	
2.		Demonstrate the Stream Editor Sed	K2	2	Lecture with demonstration	PPT
3.		Demonstrate The AWK Pattern Scanning and processing Language	K2	3	Lecture with demonstration	PPT
Total Hours				07 Hours		

UNIT – V

S#	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1.	CO5	Explain Process and its structure	K2	1	Lecture with discussion	PPT
2.		Explain the Process Identifiers	K2	1	Lecture with discussion	PPT
3.		Demonstrate Process Creation – fork(),Vfork() ,exit() System calls	K3	1	Lecture with demonstrat ion	PPT
4.		Demonstrate waiting for a process – wait(), waitpid() System calls	K3	1	Lecture with demonstrat ion	PPT
5.		Explain Zombie Process	K2	1	Lecture with discussion	PPT
6.		Demonstration of exec system call.	K3	1	Lecture with demonstrat ion	PPT
Total Hours				06 Hours		

UNIT – VI

S #	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1.	CO6	Explain Signal and signal function	K2	1	Lecture with discussion	PPT
2.		Explain unreliable signals	K2	1	Lecture with	PPT

					discussion	
3.		Explain Interrupted system calls	K2	1	Lecture with discussion	PPT
4.		Explain kill and raise Functions	K2	2	Lecture with demonstration	PPT
5.		Explain alarm, pause functions, abort, sleep functions	K2	3	Lecture with demonstration	PPT
Total Hours				08Hours		

Total No. of classes: 58

Technical Skills-III

Academic Year: 2021-22

Year/ Semester: V

Name of the Course: Technical Skills-III

Programme: B.Tech

Section: A,B,C& D

Course Code:: V18CST62

LESSON PLAN

COURSE OUTCOMES (Along with Knowledge Level):

After completion of this course, the students will be able to:

CO	Course Outcomes	Knowledge Level
1	Apply fundamental data structures like List, Stack to solve real work problems in linear time i.e. $O(n)$.	K3
2	Make use of advanced data structures like queue, to solve complex problems in linear time , logarithmic time i.e. $O(n)$ or $O(n \log n)$.	K3
3	Develop programs to solve problems by with the help of searching and sorting techniques.	K3
4	Analyze linked list by comparing with Array List and develop programs to solve optimization Problems.	K4
5	Experiment with types of Linked List to solve complex combinatorial problems.	K3
6	Develop programs to solve complex problems by using combination of stack, Queue and List.	K3

References:

1. Introduction to Algorithms, Second Edition, Thomas H. Cormen Charles E. Leiserson.
2. Data Structures and Algorithms Made Easy: Narasimha Karumanchi .
3. The Algorithm Design Manual, Springer series, Steven Skiena.

Targeted Proficiency and Attainment Levels (for each course Outcome):

Cos		C01	C02	C03	C04	C05	C06
Targeted Proficiency Level		60	60	60	60	60	60
Targeted level of Attainment	Level 3	60	60	60	60	60	60
	Level 2	50	50	50	50	50	50
	Level 1	40	40	40	40	40	40

Lecture Plan:

S#	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours required	Pedagogy	Teaching aids
1	CO 1	Dissemination of Department Vision, Mission, PEOs, POs, PSOs	-	-	-	
		Demonstrate ArrayList Operations.	K3	2	Lecture With Discussion	ICT
2		Illustrate Problem solving using ArrayList.	K3	6	Lecture With Discussion	ICT

08

S.No	Course Outcome	Intended Learning Outcomes(ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 2	Demonstrate LinkedList Operations.	K3	2	Lecture with Discussion	ICT
2		Illustrate Problem solving using LinkedList.	K3	6	Lecture with Discussion	ICT

08

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	Demonstrate Stack Operations.	K3	2	Lecture with Discussion	ICT
2		Illustrate Problem solving using Stack.	K3	10	Lecture with Discussion	ICT

12

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 4	Demonstrate Queue Operations.	K3	2	Lecture with Discussion	ICT
2		Illustrate Problem solving using Queue.	K4	8	Lecture with Discussion	ICT

10

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 5	Demonstrate Searching Techniques.	K3	2	Lecture with Discussion	ICT

2		Illustrate Problem solving using Searching.	K3	8	Lecture with Discussion	ICT
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10

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 6	Demonstrate Sorting Techniques.	K3	2	Lecture with Discussion	ICT
2		Illustrate Problem solving using Sorting.	K3	8	Lecture with Discussion	ICT

10

Total No. of Classes: 58

Professional Communication Skills - III

Academic Year: 2021-22

Programme: B.Tech

Year/ Semester: V

Section: A,B,C& D

Name of the Course: Professional Communication Skills - III Course Code::V18ENT03

LESSON PLAN

COURSE OUTCOMES (Along with Knowledge Level):

After completion of this course, the students will be able to:

CO1: Distinguish the subtle meanings of various words in different contexts, recognize similar words as well as words with contrast meanings and use them appropriately.[K2]

CO2: Interpret the passage using different strategies and answer the questions with ease.[K3]

CO3: Compare different pairs of words and draw analogy between the words. Choose an appropriate word to make a sentence meaningful.[k4]

CO4: Recognize the easiest and best possible way of solving problem in the area of Number and Letter Series, Analogy, Classification, Coding & Decoding Symbols, Ranking and Analytical Reasoning.[K1]

CO5: Investigate the different types of logics involved in Mirror and water Images, Logical Reasoning & Arithmetical Reasoning.[K4]

CO6: Find the common traps in the questions and errors likely to be made from the concepts of Blood Relations, Directions, Average, Clock and Calendar, Data Sufficiency, Permutations-Combinations and Probability.[K3]

Text Books:

T1: PIC-VOC

T2: VERBAL ABILITY-2

Reference Books:

- Sujani,T. ,et.al, 2016, Pic Voc, Sri Vasavi Engineering College.
- RAYMOND MURPHY, 2007, Essential Grammar in Use, Cambridge University press, 3rd University.
- Thomson, A.J. , 1997, A Practical English Grammar, Oxford Publications
- Alex, K. , 2010, Soft Skills, S.Chand Publications.
- Barrons, 2014, GRE-Barons, Galgotia Publications.
- Mohammed Muneer, 2004, CAT , Tata McGraw- Hill Education
- Helio Fred Garcia, 2012, The power of Communication, FT Press/Pearson Publications

- Work book-1 on Aptitude prepared by Training & Placement Cell, Sri Vasavi Engineering College.
- Magical Book on Quicker Maths-Tyra
- Practice Book on Quicker Maths-Kundan& Tyra
- R.S.Agarwal-Sultan Chand Publications
- R.S.Agarwal-Non Verbal Reasoning

Hyperlinks

1. <https://www.indiabix.com/>
2. <https://www.campusgate.co.in/>
3. <https://www.question paper.org/>

Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):

Course Outcome	Targeted Proficiency Level (% of Marks)	Targeted level of Attainment (% Students)
1	50	60
2	50	60
3	50	60
4	50	60
5	50	60
6	50	60

CO 1

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 1	To Make students understand vocabulary and its contextual usage.	K2	1	Lecture	PPT/A.V
2		Can guess the meaning of the word through the root.	K2	1	Discussion	A.V
3		Learn Synonyms-Antonyms and understand the subtle meanings of the group words.	K2	1	Lecture	PPT/A.V
4		Students learn and use new words	K2	1	Lecture	A.V
5		Can arrange sentences in	K2	1	Lecture	PPT/A.V

		sequence cohesively as per sequence				
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CO 2

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 2	Students can complete the sentence in a meaningful way.	K3	1	Lecture	PPT/A.V
2		The Format and significance of official mail is learned.	K3	1	Lecture	PPT/A.V
3		Can identify the topic sentence and construct a cohesive paragraph	K3	1	Lecture	PPT/A.V

CO 3

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	Can draw analogy between the words of different categories.	K4	1	Lecture	PPT/A.V
2		Comprehend the passage using different strategies.	K4	1	Lecture	PPT/A.V

CO 4

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 4	Students can Identify the Next letter or Number in a correct Relation.	K1	1	Lecture	PPT/A.V
2		Students can justify the relation between words and Numbers.	K1	1	Lecture	PPT/A.V
3		Students can identifying different one from group of terms.	K1	1	Lecture	PPT/A.V
4		Students can Describe their Rank in a class or in a Competition.	K1	1	Lecture	PPT/A.V

CO 5

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 5	Can Classify the ages in a family members/ Explain the relation between numbers.	K4	1	Lecture	PPT/A.V
2		Calculate the Actual time in Mirror and Water/ Classify the Images.	K4	1	Lecture	PPT/A.V
3		Differentiate the logic behind the conclusions.	K4	1	Lecture	PPT/A.V

4		Students can Explain the logic for given problem.	K4	1	Lecture	PPT/A.V
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CO 6

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 6	Students can choose the correct relation between the persons.	K3	1	Lecture	PPT/A.V
2		Students can show the correct direction.	K1	1	Lecture	PPT/A.V
3		Students can calculate the Average of data.	K1	1	Lecture	PPT/A.V
4		Students can Relate the correct day for given date and angle between two hands of a clock.	K1	2	Lecture	PPT/A.V
5		Students can Intercept data.	K1	1	Lecture	PPT/A.V
6		Students can report theReal Time Scenarios possibility..	K1	2	Lecture	PPT/A.V

Total No. of Classes:16

Advanced Data Structure

Academic Year: 2021-22

Year/ Semester: V

Name of the Course: Advanced Data Structure

Programme: B.Tech

Section: A,B,C& D

Course Code::V18CST16

LESSON PLAN

COURSE OUTCOMES (Along with Knowledge Level):

S.No.	CO No.	Course Outcome	BTL
7.	CO1	Explain external sorting method	K2
8.	CO2	Discuss pattern matching Algorithms	K2
9.	CO3	Illustrate various hash functions with appropriate examples	K3
10.	CO4	Illustrate various priority queues with appropriate examples	K3
11.	CO5	Construct self-balanced tree with appropriate examples	K3
12.	CO6	Discuss Multiway search trees	K2

Text Books:

1. Data Structures, Algorithms and Applications in C++; SartajSahni; UniverstiyPress ; 2 nd Edition.
2. Introduction to Algorithms By Thomas H Cormen, Charless E leiseron, Ronald L Rivest and Clifford Stein PHI publication Third Edition (UNIT – II)

Reference Books:

1. Data Structures, a Pseudocode Approach, Richard F Gilberg, BehrouzAForouzan, Cengage.
2. An Introduction to Data Structures with applications By Jean Paul Tremblay and Paul G Sorenson Tata McGraw Hill Second Edition
3. Fundamentals of Data Structures and algorithms by C V Sastry, Rakesh Nayak, Ch. Raja Ramesh, IK Publications, new Delhi.

Targeted Proficiency and attainment Levels (for each Course Outcome):

Cos	CO1	CO2	CO3	CO4	CO5	CO6
Targeted Proficiency Level	60	60	60	60	60	60
Targeted level of Attainment	Level 3	60	60	60	60	60
	Level 2	55	55	55	55	55
	Level 1	50	50	50	50	50

Lecture Plan:

S#	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO1	Define Data Structure and Sorting	K1	1	Lecture	BB/ICT
2		Describe External sorting and k-way merge sort	K2	1	Lecture	BB/ICT

3		Explain Time Complexity of External sorting	K2	1	Lecture	BB/ICT
4		Discuss about Buffer Handling and Parallel operation	K2	1	Lecture	BB/ICT
5		Discuss Run Generation	K2	1	Lecture	BB/ICT
6		Explain Optimal Merging of Runs	K2	2	Lecture+ Discussion	BB/ICT
7		Demonstrate the example buffer handling and parallel operations	K2	1	Lecture	BB/ICT

1	CO2	Define Strings and matching algorithms	K1	1	Lecture	BB/ICT
2		Explain Navi string matching algorithms	K2	2	Lecture	BB/ICT
3		Explain Robin-krap algorithm	K2	1	Lecture+ Discussion	BB/ICT
4		Demonstrate the String-Matching algorithm using finite automata	K2	2	Lecture+ Discussion	BB/ICT
5		Describe about Knuth Morris Pratt algorithm	K2	2	Lecture+ Discussion	BB/ICT

1	CO3	Define List and hashing	K1	1	Lecture	BB/ICT
2		Describe Dictionaries in ADT in Linear List representation	K2	2	Lecture+ Discussion	BB/ICT
3		Demonstrate about Skip List representation Ideal case Insertion and deletion	K2	2	Lecture	BB/ICT
4		Illustrate about Assigning Level	K3	1	Lecture	BB/ICT
5		Construct struct skip node and class skip list	K3	1	Lecture	BB/ICT
6		Explain about complexity of skip List methods	K2	1	Lecture	BB/ICT
7		Explain Ideal hashing	K2	1	Lecture	BB/ICT
8		Demonstrate Hash functions and tables	K3	2	Lecture+ Discussion	BB/ICT
9		Demonstrate Linear probing	K3	2	Lecture	BB/ICT
10		Illustrate Hashing with Chains	K3	2	Lecture	BB/ICT

1	CO4	Define Priority Queues	K1	1	Lecture	BB/ICT
2		Demonstrate applications of priority queues	K2	2	Lecture+ Discussion	BB/ICT
3		Demonstrate about Heap sort	K3	1	Lecture	BB/ICT
4		Demonstrate about Huffman Codes	K3	1	Lecture	BB/ICT

1	CO 5	Define Binary Search Tree.	K1	1	Lecture	BB/ICT
2		Explain about AVL Trees	K2	1	Lecture	BB/ICT
3		Describe about Red-Black Trees	K2	1	Lecture	BB/ICT
4		Demonstrate Searching Red-Black Trees	K3	1	Lecture+ Discussion	BB/ICT
5		Demonstrate Insertion into Red-Black Trees	K3	1	Lecture+ Discussion	BB/ICT
6		Demonstrate deletion from a Red-Black Tree	K3	1	Lecture+ Discussion	BB/ICT
7		Demonstrate Joining Red-Black Trees	K3	1	Lecture+ Discussion	BB/ICT
8		Explain about Splitting Red-Black Tree	K2	1	Lecture	BB/ICT
9		Explain Splay Trees	K2	1	Lecture	BB/ICT
10		Explain Splay Tree Operation	K2	1	Lecture	BB/ICT
11		Explain Amortized Complexity	K2	1	Lecture	BB/ICT

1	CO6	Define Multiway Search Trees	K1	1	Lecture	BB/ICT
2		Describe M-Way Search Trees	K2	1	Lecture	BB/ICT
3		Explain Properties M-way Search Tree	K2	1	Lecture	BB/ICT
4		Discuss about B-Trees	K2	1	Lecture	BB/ICT
5		Explain Insert Node into B-Tree	K2	1	Lecture+ Discussion	BB/ICT
6		Explain Deletion Node into B-Tree	K2	1	Lecture+ Discussion	BB/ICT

Database Management System Lab

Academic Year: 2021-22

Year/ Semester: V

Name of the Course: Database Management System Lab

Programme: B.Tech

Section: A,B,C& D

Course Code::V18CSL06 / 308

LESSON PLAN

COURSE OUTCOMES (Along with Knowledge Level):

S.No.	CO No.	Course Outcome	BTL
1.	V18CSL06.1	Build SQL Queries and Constraints	K3
2.	V18CSL06.2	Experiment with various Database Indexing Techniques.	K3
3.	V18CSL06.3	Construct PL/SQL Cursors and Exceptions	K3
4.	V18CSL06.4	Develop application programs using PL/SQL	K3
5.	V18CSL06.5	Develop PL/SQL Functions, Procedures, Packages	K3
6.	V18CSL06.6	Apply projections and aggregation on collection of MongoDB database	K3

Text Books:

1. Oracle Database 11g the Complete Reference by Oracle Press, Kevin Loney
2. Database Systems Using Oracle, Nilesh Shah, 2nd Edition ,PHI.
3. Introduction to SQL, Rick F Vander Lans, 4th Edition, Pearson Education.

Reference Books:

1. Introduction to SQL, Rick F. Vander Lans, 4th Edition, Pearson education.
2. Oracle PL/SQL Interactive Workbook, B. Rosenzweig and E. Silvestrova, 2nd Edition, Pearson education.
3. SQL & PL/SQL for Oracle 10 g, Black Book, Dr. P. S. Deshpande, Dream Tech.

Targeted Proficiency and attainment Levels (for each Course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5	CO6
Targeted Proficiency Level		65	65	65	65	65	60
Targeted level of Attainment	Level 3	65	65	65	60	60	60
	Level 2	55	55	55	50	50	50
	Level 1	45	45	45	40	40	40

Lecture Plan: **CO1:**

S.N o.	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO1	Exp1: Develop a Query to facilitate acquaintance of Built-In Functions, String Functions, Numeric Functions, Date Functions and Conversion Functions.	K3	12	Demonstration & Experiment	PPT
2		Exp2: Develop queries using operators in SQL				
3		Exp3: Develop different queries to Retrieve and Change Data using Select, Insert, Delete, and Update				
4		Exp4: Develop different queries using Group By, Order By, and Having Clauses				
5		Exp5: Develop queries on Controlling Data(Commit, Rollback, and Save point)				
6		Exp6: Develop a Query to Build Report in SQL *PLUS				
7		Exp7: Develop queries for Creating, Dropping, and Altering Tables, Views, and Constraints				
8		Exp8: Develop queries on joins and CorelatedSubqueries				

CO2:

S.N o	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
9	CO2	Exp:1 Develop different queries on Working with Index, Sequence, Synonym, Controlling Access, and Locking Rows for Update, Creating Password and Security features PL/SQL.	K3	6	Lecture & Experiment	PPT

CO3:

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
10	CO3	Exp:1 Develop PL/SQL Code using Cursors, Exceptions and Composite Data Types	K3	3	Demonstration & Experiment	PPT

CO4:

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
11	CO4	Exp1: Develop PL/SQL Code using Basic Variable, Anchored Declarations, and Usage of Assignment Operation.	K3	9	Demonstration & Experiment	PPT
12		Exp2: Develop PL/SQL Code for Bind and Substitution Variables and Printing in PL/SQL				
13		Exp3: Develop PL/SQL block using SQL and Control Structures in PL/SQL				

CO5:

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
14	CO5	Develop PL/SQL Code using Procedures, Functions, Packages and Forms	K3	3	Demonstration & Experiment	PPT

CO6:

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
15 16 17 18	CO6	Exp1: Demonstrate MongoDB Database Exp2: Develop queries to Create and drop database and collection Exp3: Develop queries to Insert,update ,delete,query document Exp4: Develop queries for Projection, limiting records, sorting records and aggregation in MongoDB	K3	6	Demonstration & Experiment	PPT

Total no of hours: 39

Operating Systems and UNIX Lab

Academic Year: 2021-22

Year/ Semester: V

Name of the Course: Operating Systems and UNIX Lab

Programme: B.Tech

Section: A,B,C& D

Course Code::V18CSL07

LESSON PLAN

COURSE OUTCOMES (Along with Knowledge Level):

- CO1:** Illustrate CPU scheduling algorithms (K3)
CO2: Apply Bankers Algorithm for Deadlock Avoidance and Deadlock Prevention (K3)
CO3: Use Page replacement algorithms for memory management (K3)
CO4: Demonstrate the basic knowledge of Linux commands and file handling utilities by using Linux shell environment. (K3)
CO5: Experiment with the concept of shell scripting programs. (K3)
CO 6: Illustrate the process of how the parent and child relationships (K3)

Targeted Proficiency and attainment Levels (for each Course Outcome):

Course Outcome	Targeted Proficiency Level (% of Marks)	Targeted level of Attainment (% Students)
CO1	60	65
CO2	60	65
CO3	60	65
CO4	60	65
CO5	60	65
CO6	60	65

LESSON PLAN

SN O	Course Outcom e	Intended Learning Outcomes (ILO)	Knowledg e Level of ILO	No. of Hours require d	Pedagogy	Teachi ng aids
1	CO1	Simulate the following CPU scheduling algorithms: a) FCFS b) SJF c) Round Robin d) Priority	K3	3	Lecture with Experiment	Black Board
2		Implement : fork (), wait (), exec() and exit () system calls	K3	3	Lecture with Experiment	Black Board

3		Simulate Producer and Consumer problem using Semaphores	K3	3	Lecture with Discussion	Black Board
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SN O	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours required	Pedagogy	Teaching aids
1	CO2	Simulate Bankers Algorithm for Dead Lock Avoidance	K3	3	Lecture with Experiment	Black Board
2		Simulate Bankers Algorithm for Dead Lock Prevention	K3	3	Lecture with Experiment	Black Board

SN O	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours required	Pedagogy	Teaching aids
1	CO3	Simulate the following page replacement algorithms: a) FIFO b) LRU c) LRU	K3	3	Lecture with Experiment	Black Board
		Simulate Memory Management Techniques: a) MFT b) MVT	k3	3	Lecture with Experiment	Black Board
2		Simulate the following File allocation strategies: a) Sequenced b) Indexed c) Linked	K3	3	Lecture with Experiment	ICT

SN O	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours required	Pedagogy	Teaching aids
1	CO4	Study of Unix Commands: General Purpose Utilities: cal,date,man,who. Directory Handling Commands: pwd,cd,mkdir,rmdir. File Handling Utilities: cat,cp,ls,rm,mv,nl,wc Displaying Commands: head, tail Filters: cmp,comm.,diff,sort,uniq Disk Utilities: du,df Changing file permissions: chmod Changing the owner of a file: chown	K3	3	Lecture with Experiment	ICT

SN O	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours required	Pedagogy	Teaching aids
1	CO5	Shell script to list all of the directory files in a directory. Shell Script to find the factorial of a given number Shell Script to generate a Multiplication table. Shell Script to Perform arithmetic operations	K3	3	Lecture with Experiment	ICT
2		Implement an AWK script to count the number of Lines in a file that do not contain vowels. Design an awk script to find the number of characters, words and lines in a file.	K3	3	Lecture with Experiment	ICT

SN O	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours required	Pedagogy	Teachin g aids
1	CO6	Demonstration of GDB tool to understand process program. Design a C program to create a Zombie Process. Design a C program that illustrates how an orphan is created.	K3	3	Lecture with Experiment	IC T